

Professional way to prepare programming contest problems

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Team Formation

time limit per test: 1 second

memory limit per test: 256 megabytes

IUBAT is hosting the ICPC Dhaka Regional 2026. The Team "IUBAT Eksho Eksho" consisting [Tanvir](#), [Imran](#) and [Fahad](#) will not be participating as the members are graduating soon.

For that, the coaches want to form exactly n teams, the coaches have gathered n Problem Readers, n Idea Generators, and n Coders. Each person has a specific skill rating.

The teams must be formed according to the **Elite Hierarchy** rule:

- Team 1 must consist of the highest-rated Reader, Idea Generator, and Coder.
- Team 2 must consist of the second-highest rated person in each category.
- ...
- Team n must consist of the lowest-rated person in each category.

Two teams are considered different if at least one person in a team is different, even if their skill ratings are identical. Calculate the total number of ways to form these n teams modulo $10^9 + 7$.

Input

The first line contains a single integer n ($1 \leq n \leq 10^5$) — the number of teams to be formed.

Problem: **n-teams-combination**, id=534686 
 AI tips: Enabled (done)
 Owner: Tanvir Ahmed (tanvir_005)
 Note:  [click here to add note](#)
 Statements: [english](#)
 Checker: std::ncmp.cpp (0)
 Validator: [val.cpp](#) (1)
 Tests: [tests](#) (56)
 Solutions: [gemini.cpp](#) (3/3)
 Package: for revision 3
 Verification: ([start](#))

[Review problem](#)

Correction tips from AI:

- for the English statement → [review](#)

Access: OWNER, Write:3, Read:1
 Revision: 3 / 3
 Invokers: 5 / 60

[View changes](#)

<https://polygon.codeforce...005/n-teams-combination> 

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1 The second line contains n integers $R_1, R_2, \dots, R_n$ ($1 \leq R_i \leq 10^9$) — the ratings of the Problem Readers.

The third line contains n integers $I_1, I_2, \dots, I_n$ ($1 \leq I_i \leq 10^9$) — the ratings of the Idea Generators.

The fourth line contains n integers $C_1, C_2, \dots, C_n$ ($1 \leq C_i \leq 10^9$) — the ratings of the Coders.

Output

Print a single integer — the number of ways to form the n teams following the Elite Hierarchy rule, modulo $10^9 + 7$.

Example

input	
3 2 2 1 1 2 3 1 1 2	
output	
4	

Note

In the example, let the participants be identified by their role and rating:

- **Readers:** $R_A(2), R_B(2), R_C(1)$
- **Idea Generators:** $I_D(1), I_E(2), I_F(3)$
- **Coders:** $C_G(1), C_H(1), C_I(2)$

According to the Elite Hierarchy, the team slots must be filled as follows:

- **Team 1:** Needs ratings (Reader: 2, Idea: 3, Coder: 2)
- **Team 2:** Needs ratings (Reader: 2, Idea: 2, Coder: 1)
- **Team 3:** Needs ratings (Reader: 1, Idea: 1, Coder: 1)

1 There are 4 possible ways to assign these specific individuals to the teams:

1. Team 1: $\{R_A, I_F, C_I\}$, Team 2: $\{R_B, I_E, C_G\}$, Team 3: $\{R_C, I_D, C_H\}$
2. Team 1: $\{R_A, I_F, C_I\}$, Team 2: $\{R_B, I_E, C_H\}$, Team 3: $\{R_C, I_D, C_G\}$
3. Team 1: $\{R_B, I_F, C_I\}$, Team 2: $\{R_A, I_E, C_G\}$, Team 3: $\{R_C, I_D, C_H\}$
4. Team 1: $\{R_B, I_F, C_I\}$, Team 2: $\{R_A, I_E, C_H\}$, Team 3: $\{R_C, I_D, C_G\}$